

EXECUTIVE SUMMARY: CLOSED-SYSTEM THEORY

Minimal Relativistic Model • Orientation of Time, Dual Constants, & Conservation Laws

CORE SYSTEM ARCHITECTURE

The model constructs a completely closed, deterministic physical system governed by **two invariant constants** (0 at array position A and light speed c), an **Orientation of Time** vector (t), and an **Euler Transformer**. All physical parameters are variables that scale dynamically while total system mass and energy remain strictly conserved.

1. System Invariants vs. Dynamic Variables

The theory distinguishes between two absolute constants and dynamic variables across the manifold:

Component	Type	Mathematical Expression & Role
Array Position 0 (A)	Invariant Constant	Absolute origin singularity (A[0]: 0=0). Serves as the division crucible (0/0 = λ) projecting unconstrained scaling seeds.
Speed of Light (c)	Invariant Constant	Universal velocity limit setting propagation bounds and fixing the mass-energy equivalence scale (E = mc²).
Orientation of Time (t)	System Orientation	Directional vector field driving trajectory evolution, quantum state transitions, and temporal acceleration.
Fields & Geometry	Dynamic Variables	Metric tensor g _{ij} , field vectors (E, B), rotation angles (θ), vectors, color, and particle states.

2. Property Extraction & The Euler Transformer

Properties (vectors, fields, attributes) are extracted out of zero and real numbers via two mechanisms:

- Extraction out of Zero (A):** At array index 0, the indeterminate form 0/0 resolves into scalar seed $\lambda \in \mathbb{N}$, projecting spatial scale factors and field attributes without requiring pre-existing background coordinate grids.
- The Euler Transformer (e^{iθ}):** The complex exponential operator $R(\theta) = e^{i\theta} = \cos \theta + i \sin \theta$ extracts orthogonal property components from any real angle θ. Evaluating a half-period rotation across π yields $e^{i\pi} + 1 = 0$, defining the exact anti-phase boundary condition where properties balance across A.

3. Closed-System Conservation & Decoupling Dynamics

Because the universe is modeled as a strictly closed system:

- Mass & Energy Invariance:** Total system mass ($m_{total} = \int \rho_m d^3r$) and total energy ($E_{total} = \int u d^3r = m_{total} c^2$) remain strictly constant across all temporal transformations.
- Atomic Decoupling & Decay:** Atoms or composite wave structures can decay, split, or decouple into particle core states and wave envelopes. During decay, localized energy density $u(\mathbf{r}, t)$ redistributes continuously across the field, keeping total energy and mass perfectly conserved without friction or energy loss.

Summary Identity: $E_{total} = m_{total} c^2 = \text{constant} \mid A (0/0 = \lambda) \mid e^{i\pi} + 1 = 0 \mid \text{Time Vector Orientation } t$